

MTH441: Lab 3

Questions

1. Consider the `cars` dataset in base `R` which contains information on the distance a car traveled after breaking. Fit a simple linear regression model and view the diagnostic plots of the model using the `plot()` command

```
# Fit linear regression model
# automatically fits with intercept
fit <- lm(dist ~ speed, data = cars)
# view summary
plot(fit)
```

The `plot` function reports four plots:

- a. Reproduce all the four plots by yourself by extracting the appropriate residuals.
 - b. Are all the residual plots satisfactory? Explain why, and if not, explain why not.
2. Consider the `Boston` dataset in the `MASS` and using `?Boston` learn about the data. The `medv` variable is the response.
 - a. Fit a multiple linear regression model, with intercept.
 - b. Center and scale the X and the y according to our notes and refit the model on this data, without intercept.
 - c. Verify that the OLS estimator of (b) is related to the OLS estimator in (a) according to the scaling diagonal matrix we studied in class.
 - d. Repeat (b) by fitting with intercept. What is the estimate of the intercept, and why?
 3. For the `Boston` dataset:
 - a. For the model in 2(a), use `plot` to look at the diagnostic plots. Are the assumptions of the model satisfied?
 - b. Using all the residuals studied in class, identify possible outliers in the data.
 - c. Using method discussed in class, identify any leverage point(s).
 - d. Using Cook's distance, identify any influence point(s).